Mini-Project 3

1 Optimal Control of the Inverted Pendulum

Using the Simulink model from Mini-Project 2, design a linear quadratic regulator (LQR) for the inverted pendulum about the upright equilibrium. Choose the state and control weighting matrices Q and R to balance pendulum stabilization, cart motion, and control effort, and apply your controller to the nonlinear model. Test the controller from a perturbed upright position and compare its response with your pole-placement controller from Mini-Project 2. In your report, show the resulting pendulum angle and control force and briefly discuss your choice of Q and R and how it affects the controller response.